1

Total Model cost = $\sum_y AnnualCost(y)$ for all years y in the model

AnnualCost(y) = **AnnualCarrierCost(y)** + **AnnualECTCost(y)** + **AnnualStorageCost(y)**

2

AnnualCarrierCost = Σ_{ec} **CostOfCarrier (ec, y)** for all energy carriers ec.

AnnualECTCost = Σ_{ect} **ECTFixedCost (ect, y)** for all energy conversion technologies ect. *Fixed cost incurred for all installations of technology ect existing in y [annualized payment over asset life]*

AnnualStorageCost = Σ_{st} **CostOfStorage (st, y)** for all energy storage technologies st. *Fixed cost incurred for all installations of storage st existing in y [annualized payment over asset life?]*

3

CostOfCarrier(ec,y) = **ECTInputVarCost(ec,y)** + $\Sigma_{ba,bt}$ **(TotalECCostInBT_BA(ec,bt,ba) * ScaleFactor)**, *where all "balancing areas" ba and "balancing time units" bt within y and ScaleFactor is a constant to scale up the cost when bt is a "representative" time unit standing for multiple "actual" time units*

ECT-InputVarCost(ec,y) = The additional cost (beyond the cost of ec itself) of processing/using ec as an input to some ect in the year y. This accounts for costs such as the cost of storing, sizing and processing coal in a coal-based plant.

TotalECCostInBT_BA(ec,bt,ba) = **ECCostInBT_BA(ec,bt,ba)** + **ECTransitCost(ec,bt,ba)** + **CostOfUnmetDemand (ec,bt,ba)**, where CostOfUnmetDemand is not a separate output but can be computed from the output UnmetDemand

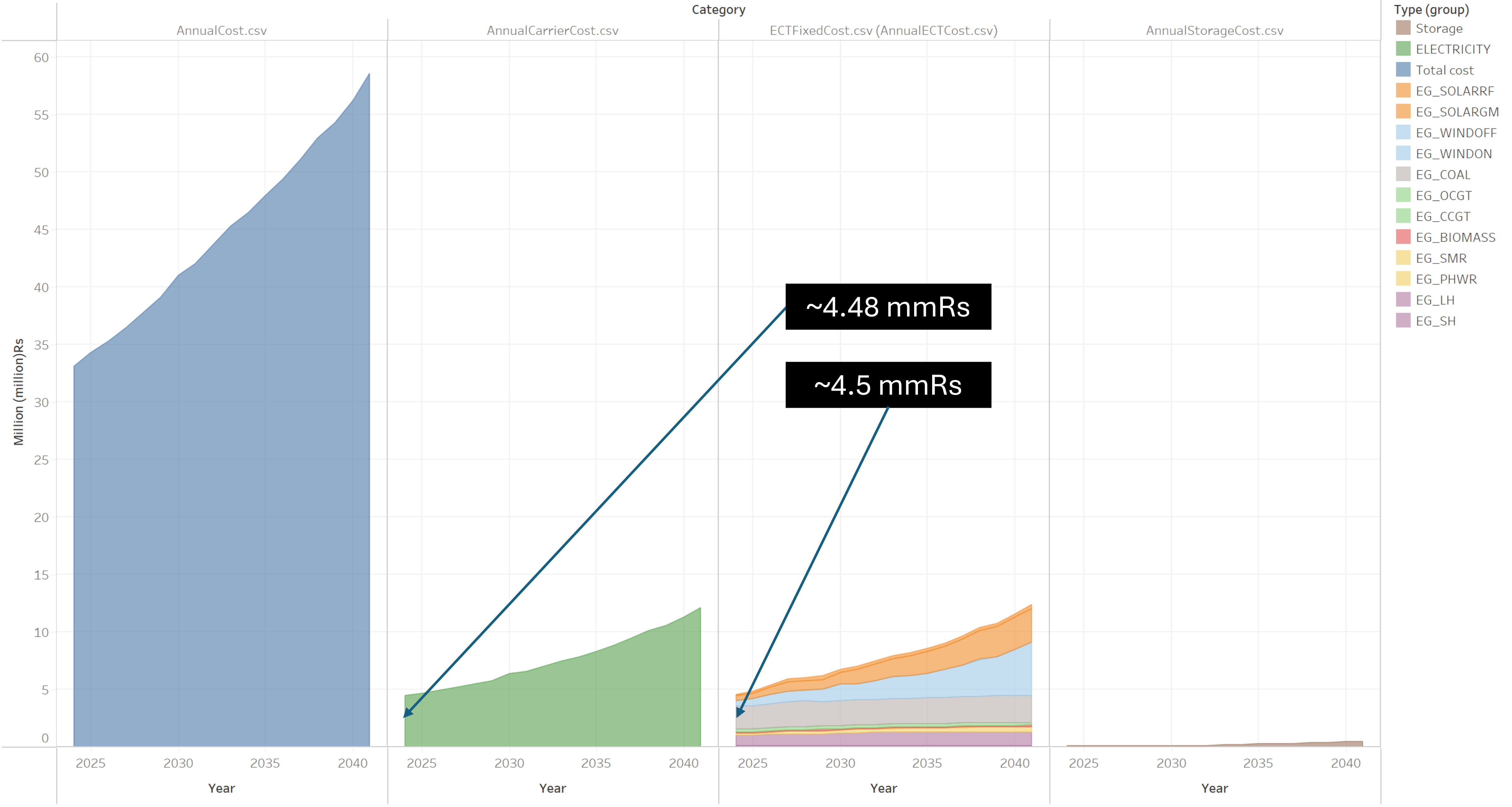
ECCostInBT_BA(ec,bt,ba) = **DomECCostInBT_BA(ec,bt,ba)** + **ImpECCostInBT_BA(ec,bt,ba)** where ImpECCostInBT_BA(ec,bt,ba) is the cost of the imported ec including taxes etc., for both primary and derived carriers, where the quantity of ec is again that imported for any purpose (end-use, input to ect or for storage)

ECTransitCost (ec,bt,ba) = The cost of transferring ec to ba from any geography ba' (including ba itself) in bt

4

DomECCostInBT_BA(ec,bt,ba) = *The quantity of ec considered here is the ec produced domestically for any purpose – to meet end-use demand, as input to some ect, or to be stored in some storage. If ec is a primary energy carrier, it is the cost of the ec domestically produced in ba-bt including taxes etc. If ec is a derived energy carrier, the only cost is the fixed tax overheads of producing the ec, because the other costs are considered elsewhere*

PIER2.0+Rumi costs – Electricity only, no electrolysis



The plot of SUM([Cost])/1e6 for Year broken down by Category. Color shows details about Type (group). The view is filtered on Type (group), which excludes 6 members.